Chunghyun Park

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Postdoctoral researcher, POSTECH Institute of AI
Citizenship: Republic of Korea

Research interests

3D visual perception for systems that act in the physical world: efficient architectures (Fast Point Transformer, PointMixer), geometric deep learning (CHOIR, RIST), open-vocabulary perception (Mosaic3D, SpaCeFormer), and active perception (Affostruction), toward 3D world action models for robot manipulation.

Education

2022 – 2026 **Pohang University of Science and Technology (POSTECH)**
Ph.D. in Artificial Intelligence
Thesis: Learning 3D Visual Perception for Geometric and Semantic Generalization
Advisor: Prof. Minsu Cho

2020 – 2022 **Pohang University of Science and Technology (POSTECH)**
M.S. in Artificial Intelligence
Thesis: Fast Point Transformer for Large-scale 3D Scene Understanding
Advisor: Prof. Jaesik Park

2014 – 2019 **Pohang University of Science and Technology (POSTECH)**
B.S. in Mechanical Engineering

Experience

Mar. 2026 – Present **POSTECH Institute of AI** (Postdoctoral researcher) – Pohang, Republic of Korea
Host: Prof. Minsu Cho
Mandatory military service until August 2027.
Working on 3D world action models and active perception for robot manipulation.

Mar. 2025 – Feb. 2026 **NVIDIA** (Research collaboration) – Santa Clara, USA (Remote)
Collaborators: Chris Choy, Jan Kautz
Published SpaCeFormer at ICML 2026, extending Mosaic3D to open-vocabulary 3D instance segmentation.

Dec. 2023 – Feb. 2025 **NVIDIA** (Research intern) – Taipei, Taiwan (Remote)
Hosts: Jaesung Choe, Yu-Chiang Frank Wang
Published Mosaic3D at CVPR 2025, a foundation dataset and model for open-vocabulary 3D segmentation.

Publications

** indicates equal contribution.*

2026 **Affogato: Learning Open-Vocabulary Affordance Grounding with Automatic Data Generation at Scale**
Junha Lee*, Eunha Park*, Chunghyun Park, Dahyun Kang, Minsu Cho
European Conference on Computer Vision (ECCV)

2026 **SpaCeFormer: Fast Proposal-Free Open-Vocabulary 3D Instance Segmentation**
Chris Choy, Junha Lee, Chunghyun Park, Minsu Cho, Jan Kautz
International Conference on Machine Learning (ICML)

2026 **Affostruction: 3D Affordance Grounding with Generative Reconstruction**
Chunghyun Park, Seunghyeon Lee, Minsu Cho
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2025 **Combinative Matching for Geometric Shape Assembly**
Nahyuk Lee*, Juhong Min*, Junhong Lee, Chunghyun Park, Minsu Cho
IEEE/CVF International Conference on Computer Vision (ICCV) Highlight

2025 **Mosaic3D: Foundation Dataset and Model for Open-Vocabulary 3D Segmentation**
Junha Lee*, Chunghyun Park*, Jaesung Choe, Yu-Chiang Frank Wang, Jan Kautz, Minsu Cho, Chris Choy
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2024 **Learning SO(3)-Invariant Semantic Correspondence via Local Shape Transform**
Chunghyun Park*, Seungwook Kim*, Jaesik Park, Minsu Cho
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2023 **Stable and Consistent Prediction of 3D Characteristic Orientation via Invariant Residual Learning**
Seungwook Kim*, Chunghyun Park*, Yoonwoo Jeong, Jaesik Park, Minsu Cho
International Conference on Machine Learning (ICML)

2022 **SeLCA: Self-Supervised Learning of Canonical Axis**
Seungwook Kim*, Yoonwoo Jeong*, Chunghyun Park*, Jaesik Park, Minsu Cho
NeurIPS Workshop on Symmetry and Geometry in Neural Representations

2022 **PointMixer: MLP-Mixer for Point Cloud Understanding**
Jaesung Choe*, Chunghyun Park*, Francois Rameau, Jaesik Park, In So Kweon
European Conference on Computer Vision (ECCV)

2022 **Fast Point Transformer**
Chunghyun Park, Yoonwoo Jeong, Minsu Cho, Jaesik Park
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2021 **Improved Classification and Localization Approach to Small Bowel Capsule Endoscopy using Convolutional Neural Network**
Yunseob Hwang*, Han Hee Lee*, Chunghyun Park, Bayu Adhi Tama, Jin Su Kim, Dae Young Cheung, Woo Chul Chung, Young-Seok Cho, Kang-Moon Lee, Myung-Gyu Choi, Seungchul Lee, Bo-In Lee
Digestive Endoscopy

Patents

2025 **Hardware Apparatus and Method for Predicting Structural Characteristic Orientation of Point Cloud of Structural Object**
Seungwook Kim*, Chunghyun Park*, Yoonwoo Jeong, Jaesik Park, Minsu Cho
U.S. Patent Application No. US 2025/0191218 A1

Awards & honors

2026 Outstanding Presentation Award, Doctoral Colloquium (KCCV)

2026 KCCV Doctoral Colloquium
Awarded to 9 selected Ph.D. graduates in computer vision across South Korea

2026 CVPR Doctoral Consortium
Awarded to 24 selected Ph.D. graduates in computer vision worldwide

2026 Bronze Prize, Best Paper Award (IPIU)

2025 POSTECHIAN Fellowship – Innovation (POSTECH)
For outstanding graduate students of the semester (₩6,000,000)

2024 CVPR Outstanding Reviewer
Top 2% of 9,872 reviewers

2023 Grand Prize, BK21 Best Paper Award (POSTECH GSAI)

2022 Qualcomm Innovation Fellowship (Qualcomm Korea)
Top AI papers by graduate students in South Korea (₩4,000,000)

2022 Silver Prize, Samsung HumanTech Paper Award (Samsung Electronics)
Top papers across 10 fields nationwide (₩7,000,000)

Talks

Aug. 2026 Embodied Vision for Real-World Perception and Action
Oral presentation, KCCV Doctoral Colloquium

Apr. 2026 Generalizable 3D Perception for Physical AI
Invited talk, Chung-Ang University
Hosts: Prof. Jongmin Lee, Prof. Sooyoung Lee

Mar. 2024 Equivariant Networks for 3D Vision
Guest lecture, CSED703J Symmetry and Equivariant Learning

Nov. 2023 Stable and Consistent Prediction of 3D Characteristic Orientation
Spotlight presentation, Korea AI Summit

Professional service

Journal Reviewer

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
International Journal of Computer Vision (IJCV)

Conference Reviewer

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
IEEE/CVF International Conference on Computer Vision (ICCV)
European Conference on Computer Vision (ECCV)
Conference on Neural Information Processing Systems (NeurIPS)
International Conference on Learning Representations (ICLR)